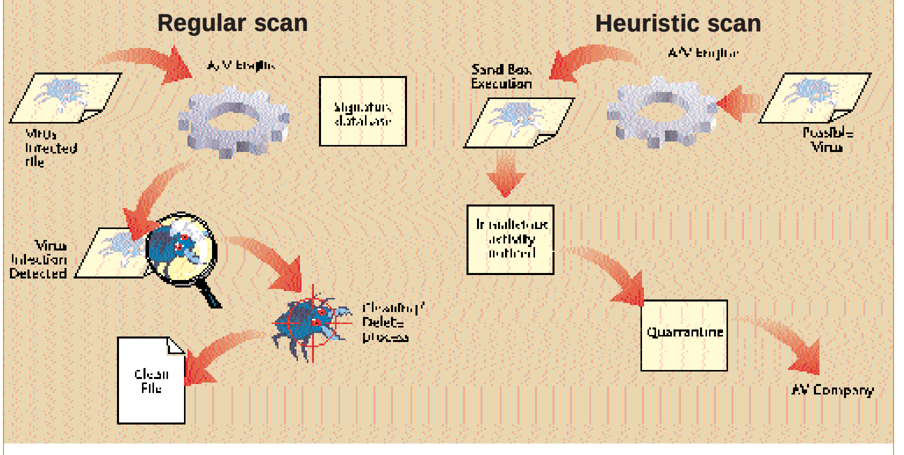
INFOGRAPHICS: Solomon Lewis

What makes Heuristic Scan better than Regular Scan



Heuristic scanning finds and quarantines even the latest, undetected virus. In Heuristic scanning, the anti-virus executes the infected file in a sandbox, checks for malicious activity, and quarantines the file if any such activity is detected. It is not just based on signatures.

database, commonly known as 'virus definitions', is nothing but a collection of 'hex data' that the anti-virus engine uses to perform searches on a file, and if found, it tries to repair the file without destroying the original data.

As viruses have numerous variants and many new viruses are developed everyday, it becomes very hard to keep track of each and every virus signature. That's why anti-virus tools started using a new method, in conjunction with the existing method of using anti-virus definitions. This method is called heuristic scanning. An anti-virus takes a piece of code and runs it in a virtual environment called a sandbox and observes its behaviour. If it finds any malicious activity, it declares the piece of code a virus and prompts the user for the necessary action to be taken. Generally, if the user allows, the sample code is quarantined and sent to the anti-virus company for classification and categorisation. However, just installing an anti-virus is not enough.

Here are a few steps that are essential even with anti-virus software installed.

- Create an emergency disk which will contain the DOS version of the anti-virus software along with the ability to clean viruses. The emergency disk should be a bootable CD or a bootable floppy. Write-protect the floppy and keep it in a safe store.
- Set your anti-virus software to automatically download the updates. If automatic updates are not available, check for updates at least once a week.
- Set up the anti-virus software to automatically scan downloads and e-mail.
- Set up the anti-virus software to automatically scan any removable media, especially if you copy files from the removable media.
- Do not disable the anti-virus software that is residing in the taskbar and monitoring all files.
- Schedule a virus scan once every week.
- In the settings for the anti-virus software, enable the highest protection possible. Keep it to scan all files by default (as opposed to only program files).
- Set the setting to automatically clean the file if a virus is found and, if cleaning is not possible, to isolate the file or delete it. This will allow you to schedule automated scanning without your intervention in case a virus is found.
- Do not disable logging for any virus scans. Make it a point to view the logs after every scheduled scan or a full-system scan.

Under siege

How exactly does a virus spread? Whenever a computer starts or boots up, it first accesses the boot sector and then starts loading the operating system. Boot sector viruses infect the boot sector by attaching themselves to this area. As soon as the computer starts the virus lodges itself in the memory and from there on slowly extends its presence by entering other areas of the computer. Once it enters the system, it tries to enter other computers, for which it needs a medium whereby it can carry itself to other PCs. The normal medium of transfer can be e-mail, Web sites, instant messengers and P2P networks, storage media (floppies, zip cartridges, CDs, hard disks, etc) and also through LAN connections.

E-mail

If sentences like 'I love you' and 'Hi, I need your advice' have become commonplace among techies, you know what to blame—e-mail. The most common Internet function is also the most dangerous one as evident from the way worms such as 'Iloveyou' or Sircam spread throughout the world. As per a *PC Magazine* study, almost 80 per cent of all viruses spread through e-mail, that is e-mail attachments, and especially

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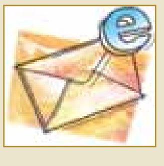
E-mail Safety Rules



- Don't open an attachment from a stranger.
- Think before you click on an attachment. Just because the attachment doesn't have a .exe extension or is not a Word file doesn't mean it cannot be a virus. Compressed files and executable files can have viruses within them renamed with .gif or .jpg extensions. These viruses get executed the moment you open the file. As a principle, scan all attachments irrespective of file type before opening.
- Look out for suspicious subject lines—'I Love You' and 'Hi, I need your advice' are already famous. The next one might be 'Mama wants you back!'
- Use e-mail filters to block senders or remove e-mail from the server directly.
- Use an anti-virus software to directly scan e-mail before downloading.
- Apply security updates to your e-mail client and browser on a regular basis. You can keep track of bugs in the software and the available updates at ***www.securityfocus.com.***
- Disable options for ActiveX scripts.
- Work without a preview pane.
- Keep your address book to a minimum. That way, even if you get infected, you spread the virus to a few people.
- Look out for messages that start with 'An ActiveX script prevents the message from being disabled properly.'